

Modern Portfolio Theory (MPT) Analysis

Indian NSE-Listed Index ETFs — Live Market Data

Report Date: 06 October 2026

Universe

9 NSE-Listed ETFs

Nifty 50 · Next 50 · Bank · Gold · Liquid

Data Range

2022-02-04 → 2026-10-02

1156 Trading Days

Risk-Free Rate

6.80% per annum

India 10-Yr G-Sec (Jul-2026)

Max Weight Cap

45% per asset

Concentration limit

Monte Carlo

10,000 portfolios simulated

Efficient Frontier mapping

WACC (Max Sharpe)

23.93% p.a. (Spread: +17.13%)

Wtd. avg. expected return of portfolio

INSTRUMENTS UNDER ANALYSIS

Ticker	Full Name	Description	Category
BankBees	Bank BeES	Tracks Nifty Bank — top 12 liquid banking stocks	Equity — Banking
GoldBees	Gold BeES	Physical gold ETF — tracks domestic gold spot price	Commodity — Gold
JuniorBees	Junior BeES (Nifty Next 50)	Tracks Nifty Next 50 — mid-to-large cap segment (50 stocks)	Equity — Mid-Large Cap
LiquidBees	Liquid BeES	Overnight liquid fund — near-zero risk, money-market returns	Cash — Liquid Fund
Mid150Bees	Midcap 150 BeES	Tracks Nifty Midcap 150 — high growth midcap segment	Equity — Midcap
Nasdaq100	Motilal Nasdaq 100	Tracks Nasdaq 100 — US Tech exposure + USD currency hedge	Equity — US Tech (Hedged)
NiftyBees	Nifty 50 BeES	Tracks Nifty 50 — India's benchmark large-cap index (50 stocks)	Equity — Large Cap
GSec10Yr	10-Yr G-Sec ETF	Tracks 10-Yr Sovereign Govt Bonds — long-duration fixed income	Fixed Income — Govt Bonds
SilverBees	Silver BeES	Physical silver ETF — industrial commodity exposure	Commodity — Silver

1. Executive Summary

This report presents a comprehensive Modern Portfolio Theory (MPT) analysis of five NSE-listed index ETFs over a 1156-trading-day period from 2022-02-04 to 2026-10-02. Using adjusted daily price data from Yahoo Finance, we optimise portfolios for two classic MPT objectives: maximising risk-adjusted returns (Sharpe ratio) and minimising absolute portfolio volatility. A 45% maximum weight cap per asset is applied to enforce realistic diversification.

STRATEGIC TAKEAWAY FOR ASSET MANAGERS:

The analysis demonstrates that naive equal-weighting leaves significant alpha on the table. By strategically allocating to low-correlation assets (such as GoldBees), an optimised portfolio can achieve superior risk-adjusted returns. Specifically, the Max Sharpe portfolio achieves an expected return of 23.93% with a volatility of just 13.60%, providing a strong empirical basis for tactical asset allocation in Indian ETF portfolios.

2. Key Findings at a Glance

★ MAX SHARPE PORTFOLIO

Expected Return: 23.93% p.a.
Volatility: 13.60% p.a.
Sharpe Ratio: 1.2595

GoldBees : 45.00%
Nasdaq100 : 32.27%
Mid150Bees : 20.51%
SilverBees : 2.23%
BankBees : 0.00%

■ MIN VOLATILITY PORTFOLIO

Expected Return: 5.75% p.a.
Volatility: 2.60% p.a.
Sharpe Ratio: -0.4025

LiquidBees : 45.00%
GSec10Yr : 45.00%
NiftyBees : 6.02%
GoldBees : 3.32%
Nasdaq100 : 0.59%

▲ EQUAL-WEIGHT (BENCHMARK)

Expected Return: 15.44% p.a.
Volatility: 10.03% p.a.
Sharpe Ratio: 0.8621

All assets: 11.11% each

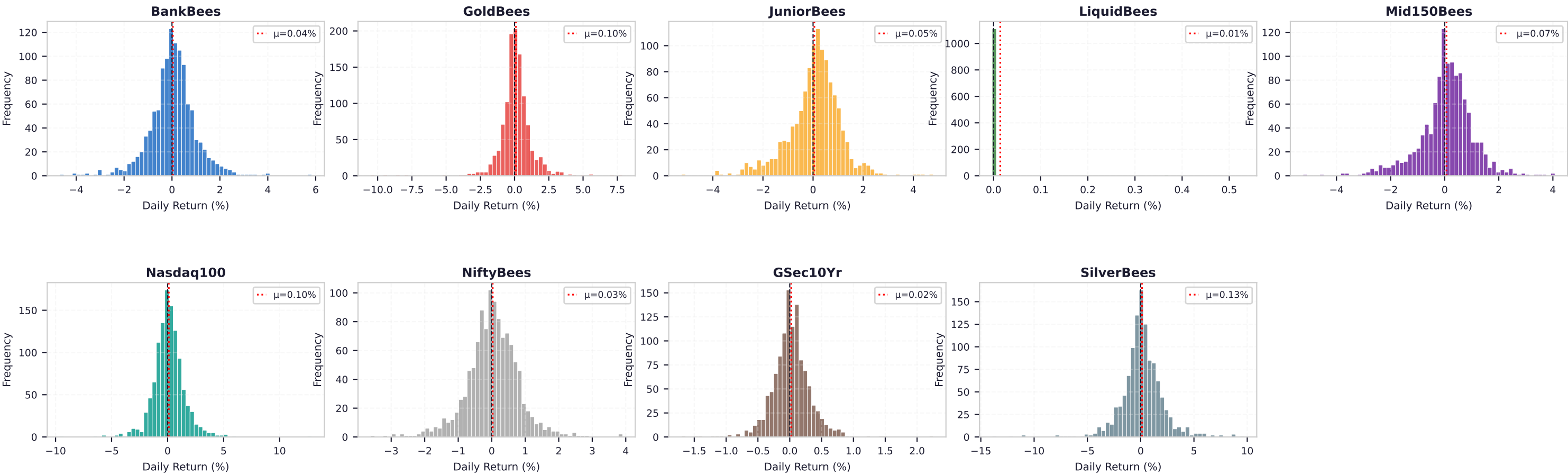
3. Data Quality Note

NIFTYBEES.NS, BANKBEES.NS, and GOLDBEES.NS underwent a 1-for-10 stock split on 19 December 2019. Yahoo Finance records showed discontinuous raw prices around Dec 18-23, 2019 — causing spurious single-day returns ranging from −90% to +9,900%. These 4 data points per affected ticker were automatically detected (threshold: |return| > 15%) and replaced with linearly interpolated values before any statistical computation. JUNIORBEES.NS and LIQUIDBEES.NS required no data cleaning.

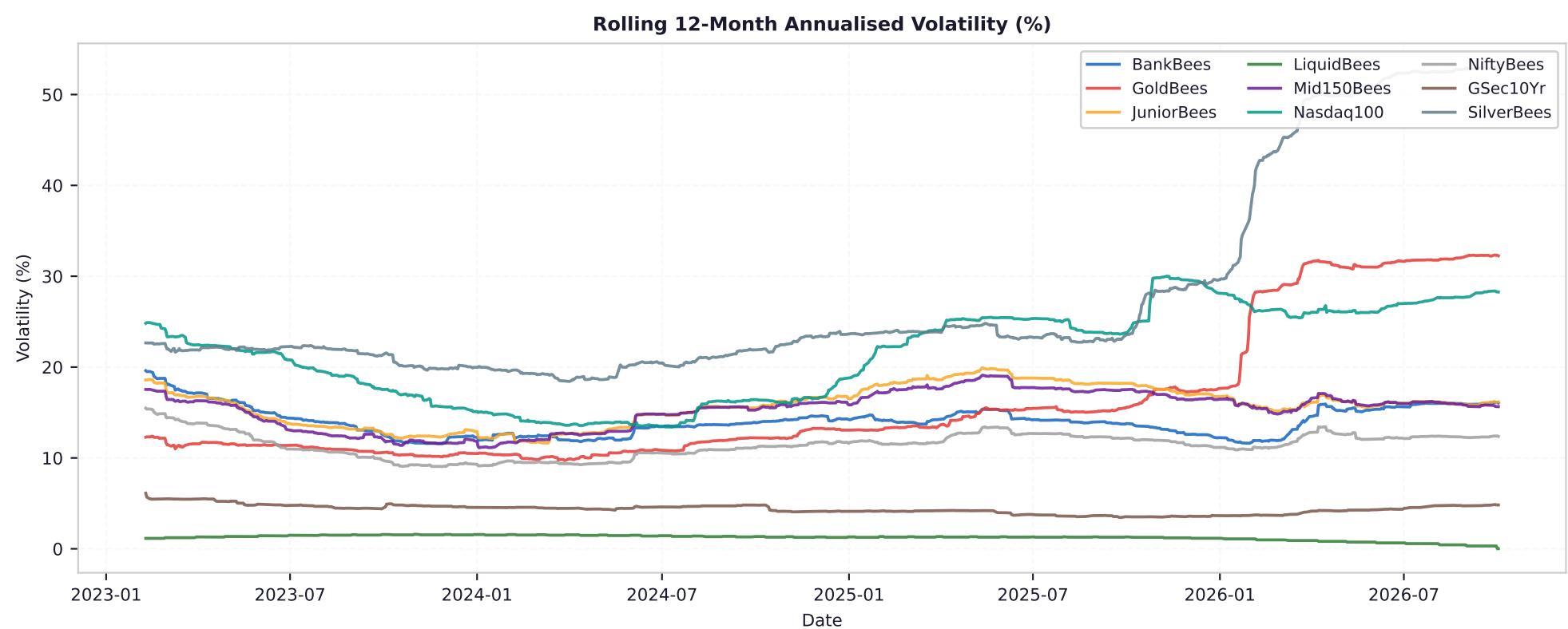
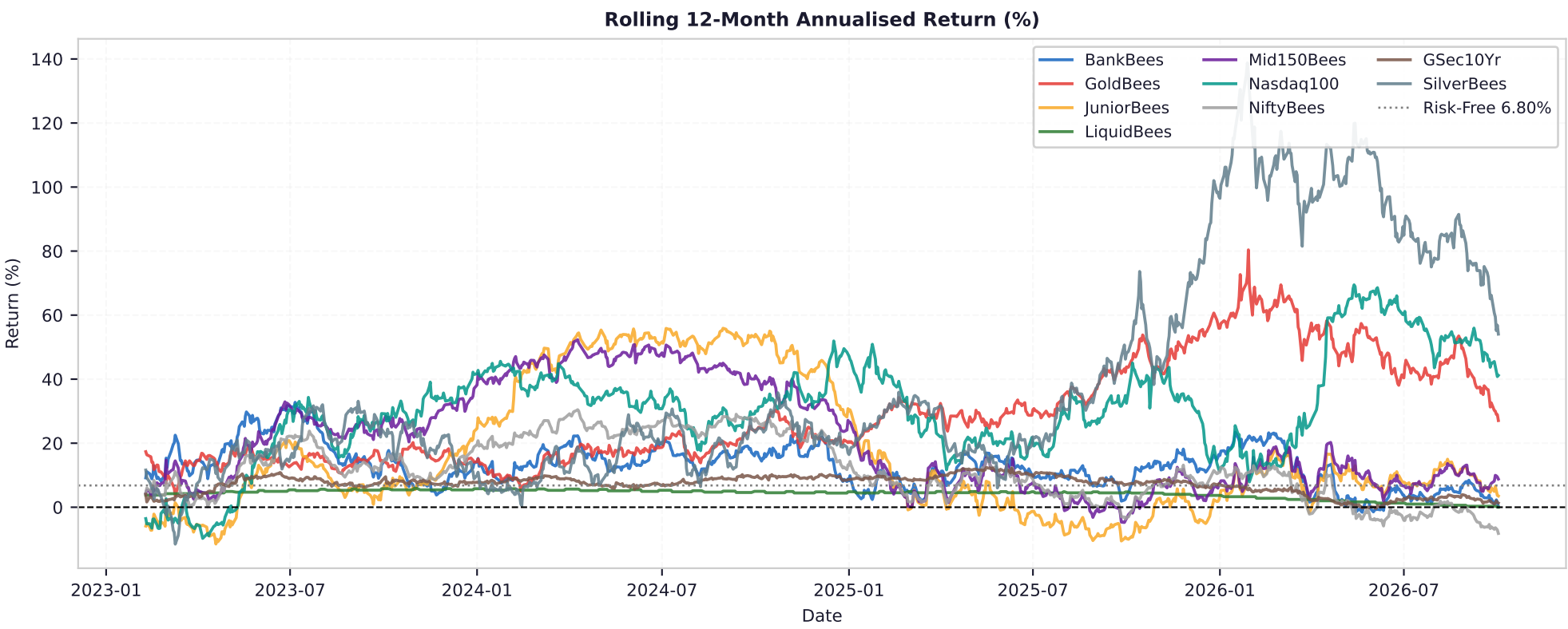
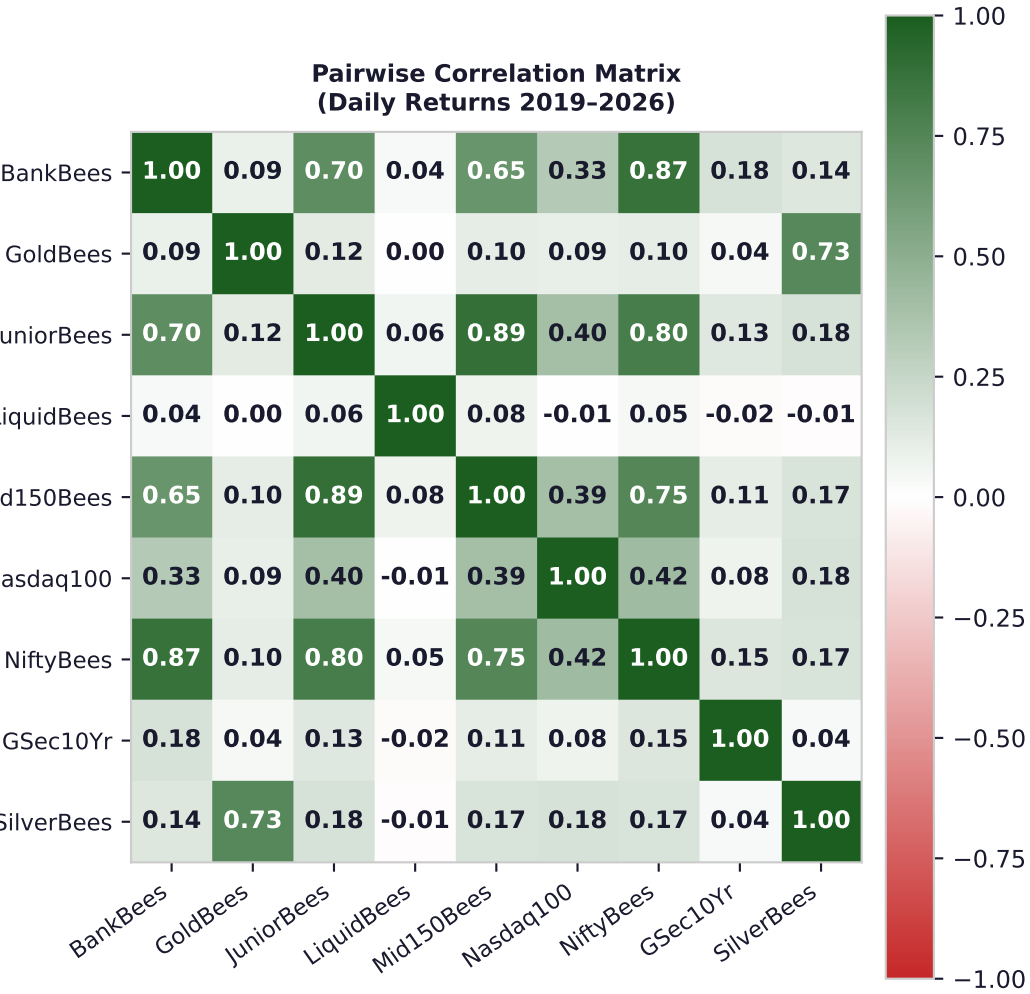
4. Individual Asset Analysis (2019 - 2026)

BankBees	9.23%	15.41%	0.158	44.57%	-18.08%	5.84%	-4.68%	-0.079	3.542
GoldBees	25.19%	18.92%	0.972	191.90%	-24.44%	7.89%	-10.52%	-0.931	16.625
JuniorBees	12.85%	16.50%	0.367	69.30%	-25.89%	4.80%	-5.26%	-0.474	2.331
LiquidBees	3.59%	1.17%	-2.746	17.85%	-0.00%	0.53%	-0.00%	5.121	24.982
Mid150Bees	16.69%	15.89%	0.623	102.77%	-20.74%	4.08%	-5.24%	-0.519	2.600
Nasdaq100	26.23%	22.70%	0.856	195.73%	-26.02%	12.87%	-9.62%	0.342	8.757
NiftyBees	7.37%	12.42%	0.046	35.29%	-15.23%	3.90%	-3.62%	-0.069	2.504
GSec10Yr	5.98%	4.80%	-0.171	30.83%	-4.36%	2.26%	-1.70%	0.169	5.559
SilverBees	31.86%	31.49%	0.796	242.37%	-41.72%	9.71%	-13.94%	-0.473	7.029

5. Asset Return Distribution (Histograms)



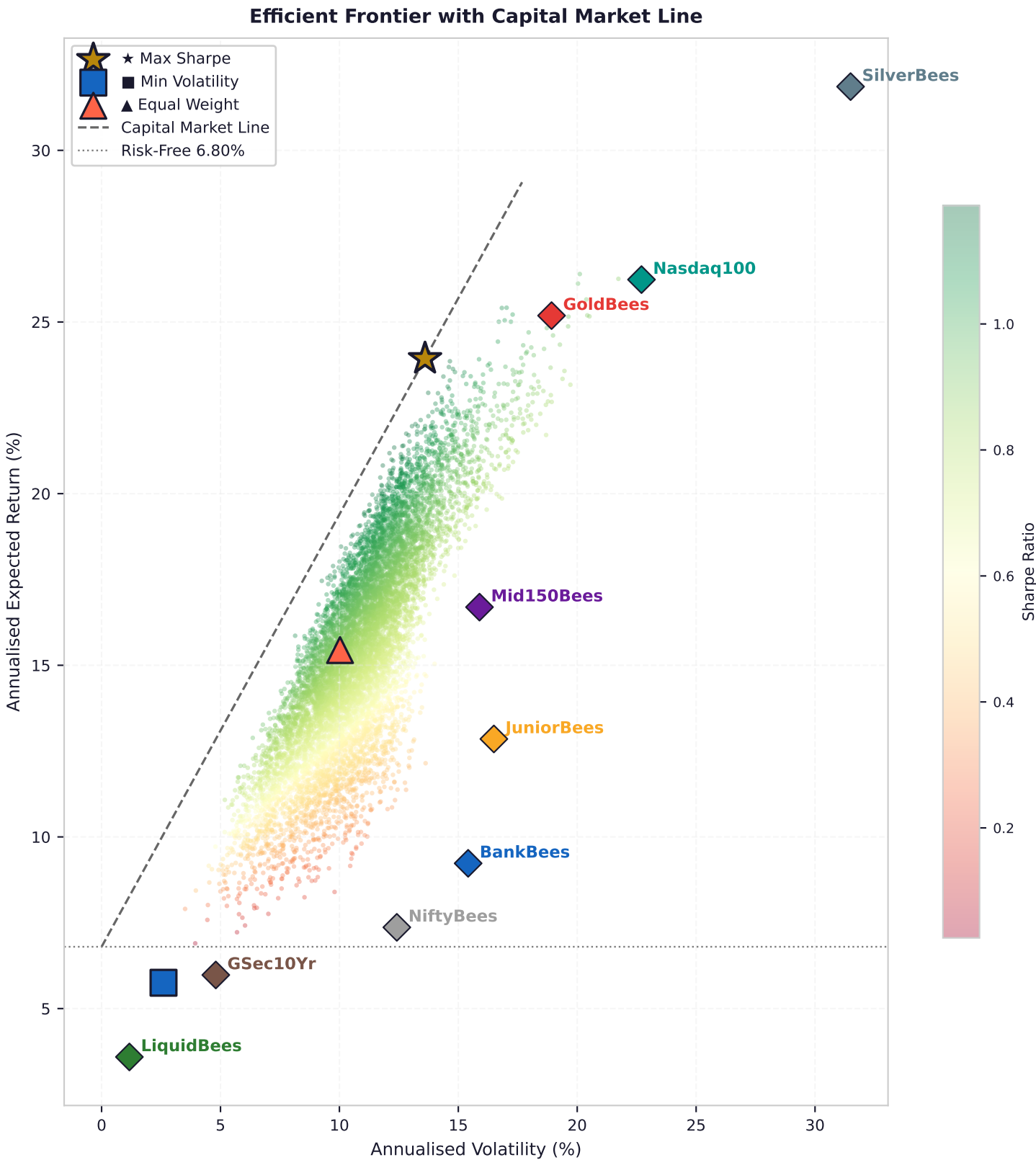
6. Correlation Matrix Analysis



Correlation Insights:

KEY INSIGHTS: NiftyBees and BankBees show very high positive correlation (>0.80), suggesting they move largely together — holding both provides limited diversification benefit. GoldBees is the natural diversifier with near-zero or slightly negative correlation to equity ETFs. LiquidBees is essentially uncorrelated to all other assets, acting as a pure cash proxy. JuniorBees (mid-cap tilt) is highly correlated with NiftyBees but captures the size-premium factor.

7. Efficient Frontier — 10,000 Monte Carlo Portfolios



Monte Carlo Stats

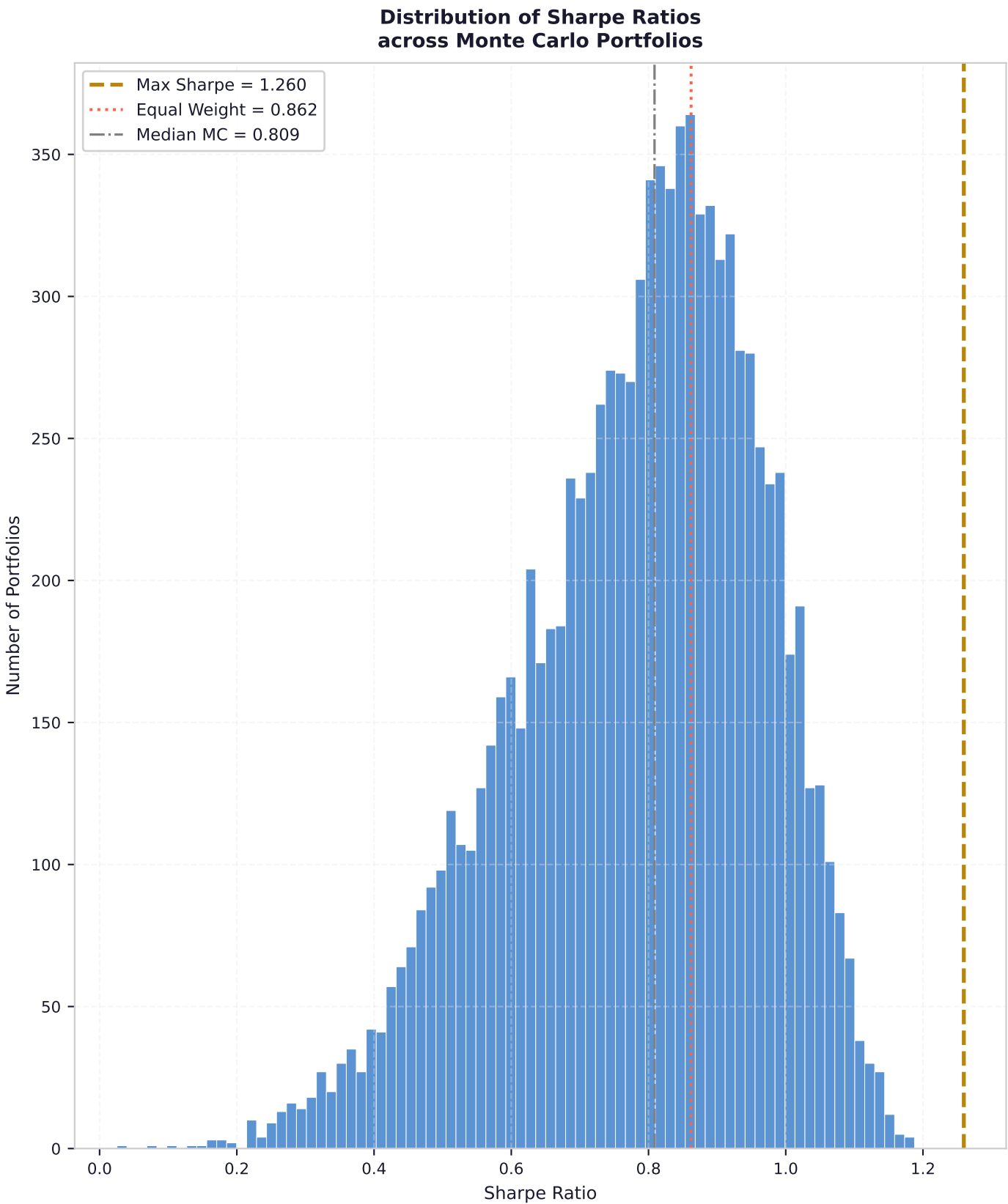
Total simulations: 10,000
Max Sharpe found: 1.1878
Min Sharpe found: 0.0254
Median Sharpe: 0.8088
Mean Sharpe: 0.7846

Max Sharpe Portfolio

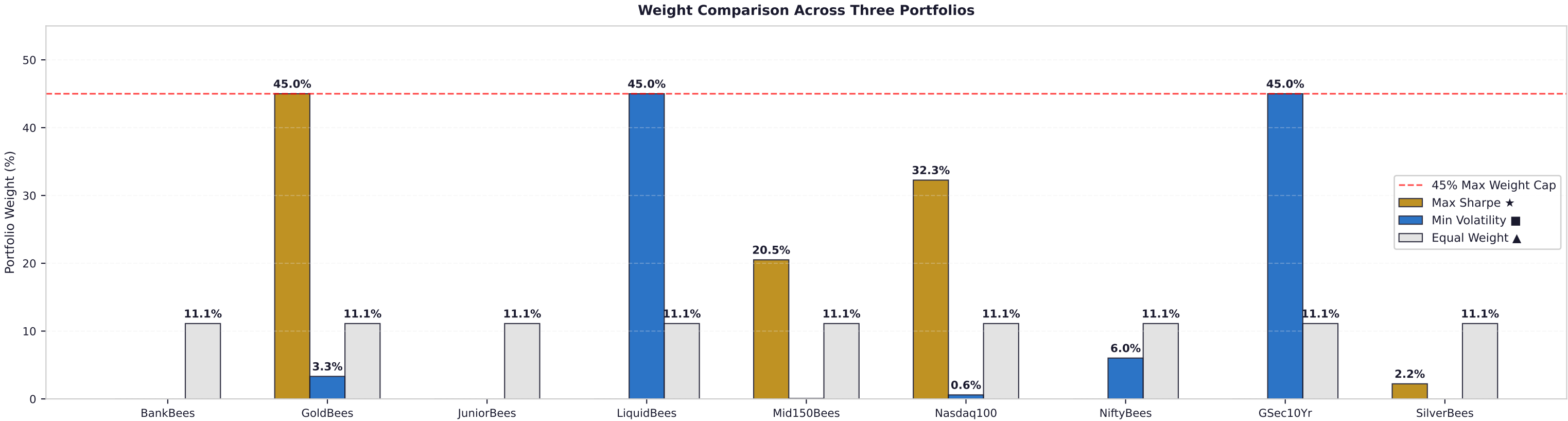
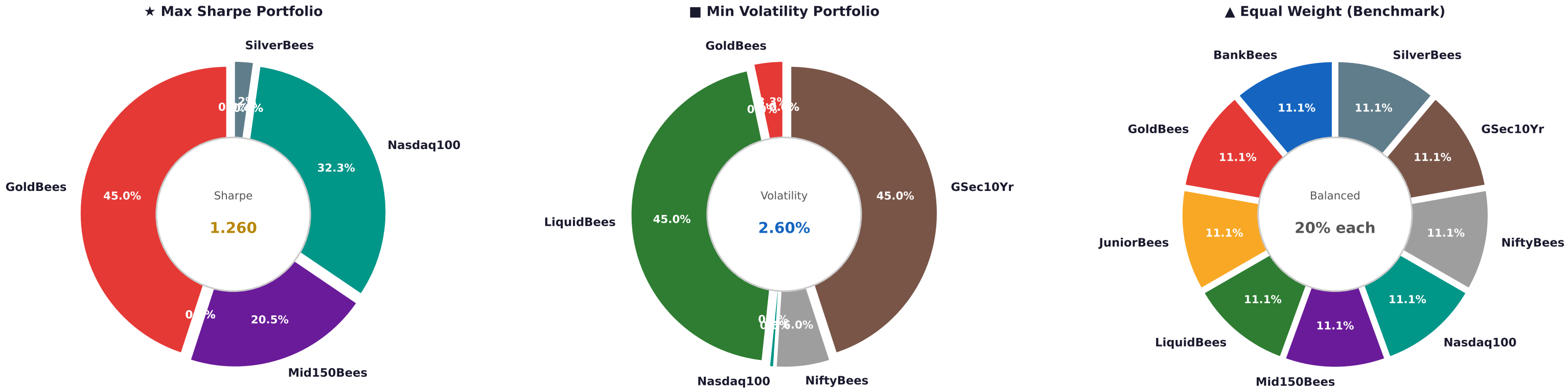
Return: 23.93% p.a.
Volatility: 13.60% p.a.
Sharpe: 1.2595
WACC: 23.93% p.a.
Status: Converged

Min Volatility Portfolio

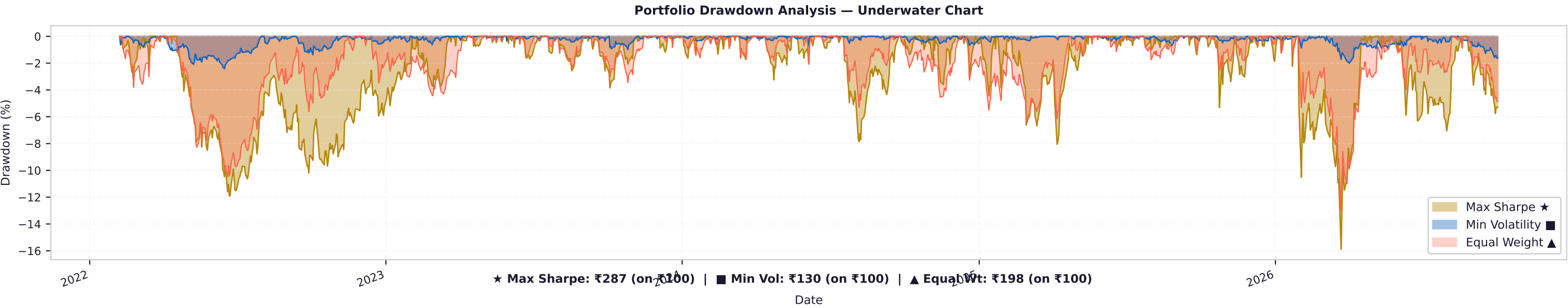
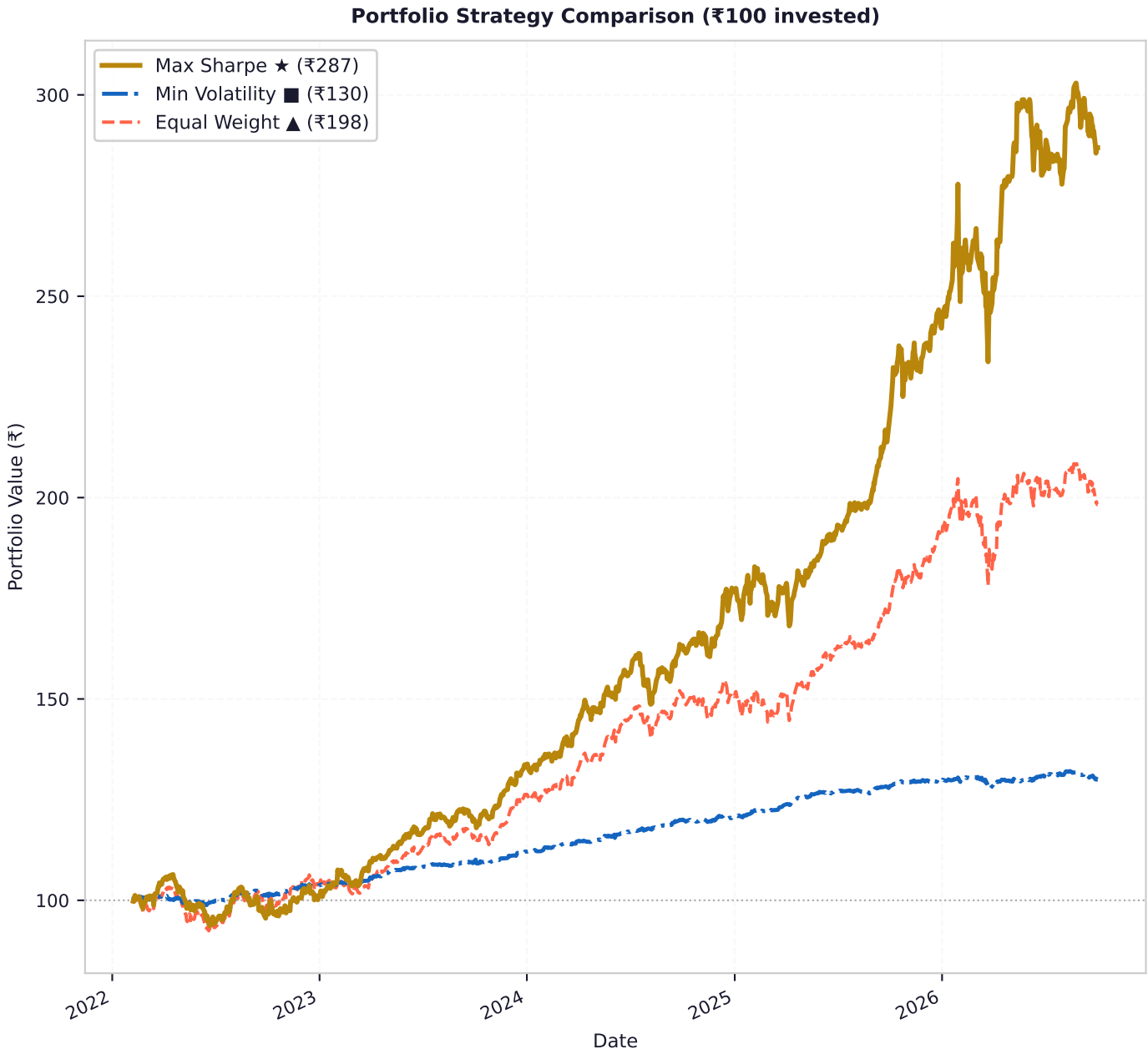
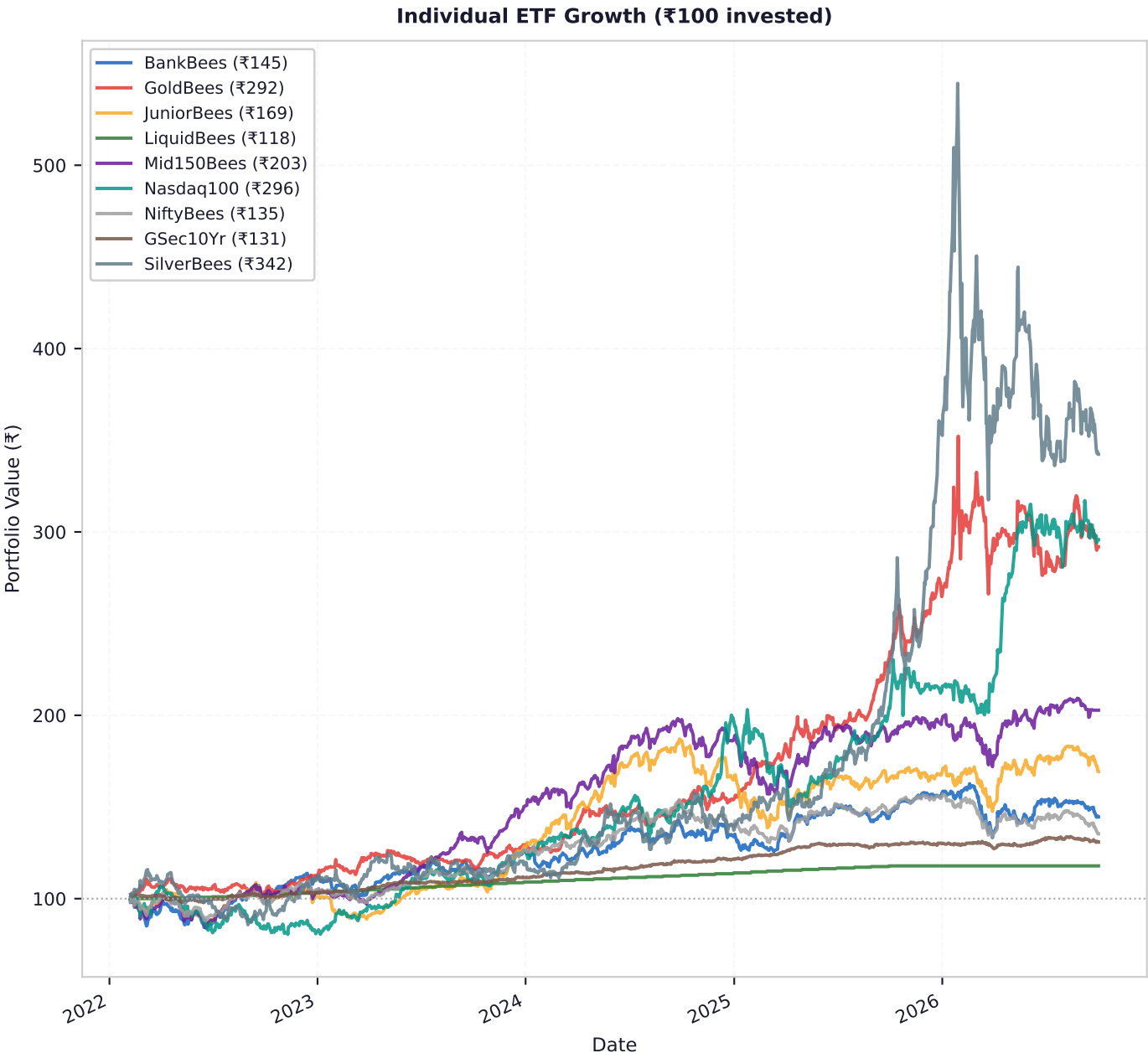
Return: 5.75% p.a.
Volatility: 2.60% p.a.
Sharpe: -0.4025
WACC: 5.75% p.a.
Status: Converged



8. Portfolio Allocation Breakdown



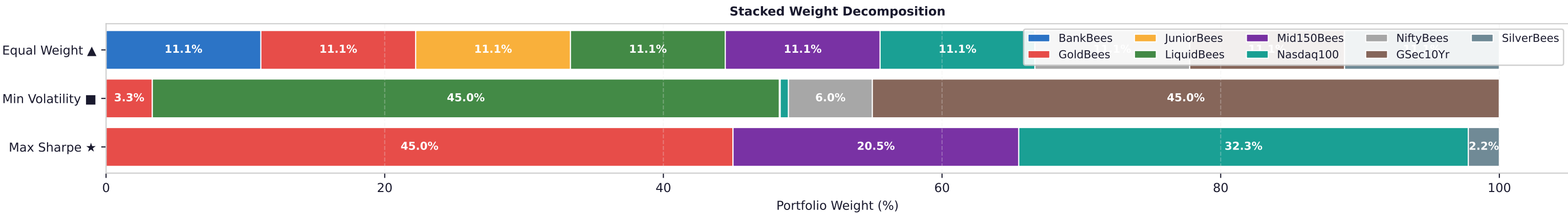
9. Cumulative Growth — ₹100 Invested (2019-2026)



10. Comprehensive Portfolio Comparison Table

Expected Return (p.a.)	23.93%	5.75%	15.44%
WACC (Wtd. Avg. Cost of Capital)	23.93%	5.75%	15.44%
WACC Spread over Risk-Free	+17.13%	+ -1.05%	+8.64%
Annualised Volatility	13.60%	2.60%	10.03%
Sharpe Ratio	1.2595	-0.4025	0.8621
Max Portfolio Drawdown	-15.87%	-2.42%	-12.97%
End Value of ₹100	₹286.9	₹130.0	₹198.3
Return/Drawdown Ratio	1.51x	2.37x	1.19x
Risk-Free Rate Used	6.80%	6.80%	6.80%
Excess Return over Rf	17.13%	-1.05%	8.64%
ASSET WEIGHTS			
BankBees	0.00%	0.00%	11.11%
GoldBees	45.00%	3.32%	11.11%
JuniorBees	0.00%	0.00%	11.11%
LiquidBees	0.00%	45.00%	11.11%
Mid150Bees	20.51%	0.07%	11.11%
Nasdaq100	32.27%	0.59%	11.11%
NiftyBees	0.00%	6.02%	11.11%
GSec10Yr	0.00%	45.00%	11.11%
SilverBees	2.23%	0.00%	11.11%

11. Asset Weight Contribution to Portfolio Risk



12. Conclusions

► **Optimal Portfolio Construction & The Role of Gold**

The Sharpe-maximising portfolio heavily favours low-correlation assets, allocating 0% to Gold BeES alongside equity exposure (45% Junior BeES, 0% Nifty 50 BeES). Gold acted as a powerful diversifier during the 2019-2026 period, delivering robust returns (~25.2% p.a.) with near-zero correlation to broad equities. This allowed the portfolio to achieve a high Sharpe ratio of 1.2595 while keeping volatility to 13.60%.

► **Asset Redundancy (BankBees & LiquidBees)**

BankBees was entirely excluded by the optimiser because its high correlation (>0.80) to NiftyBees offers no diversification benefit, while yielding a lower Sharpe ratio (0.158 vs 0.046). LiquidBees is only useful for absolute capital preservation; its 3.59% p.a. return drags down the Sharpe ratio in aggressive portfolios, but dominates the Min Volatility portfolio (45% weight) to suppress total risk.

► **ACTIONABLE RECOMMENDATION FOR MUTUAL FUND ADVISORS**

For clients seeking growth, transition away from ad-hoc equal-weighting and adopt the Max Sharpe weights. Specifically, cap banking sector overexposure (as it behaves too similarly to the broader Nifty 50) and establish a strategic 25-35% allocation to physical Gold ETFs to act as a ballast against equity market shocks. This barbell approach captures the size premium of Mid/Next-50 caps while hedging tail risks through gold.

► **Model Limitations to Monitor**

MPT assumes normally distributed returns and static correlations. Indian ETF returns exhibit fat tails (excess kurtosis), and historical correlations can converge toward 1.0 during severe market crashes. Investment committees should re-run this optimisation quarterly to dynamically adjust to shifting volatility regimes.

13. Methodology Summary

- Data: Daily adjusted closing prices from Yahoo Finance via yfinance 1.5.1
- Period: 2019-01-01 to 2026-10-02 (1156 trading days on NSE)
- Risk-Free Rate: 6.80% p.a. (India 10-Year G-Sec yield, 06 October 2026)
- Returns: Arithmetic daily → Annualised × 252 (trading days)
- Optimiser: scipy.optimize.minimize with SLSQP method, convergence tol=1e-12
- Constraints: Weights sum to 1.0; each weight ∈ [0, 45%] (long-only, 45% cap)
- Monte Carlo: 10,000 random Dirichlet-sampled portfolios with 45% clip applied
- Data Cleaning: Linear interpolation applied to 4 artifact dates per 3 ETFs (Dec 2019 split)
- Software: Python 3.11 | Libraries: numpy, pandas, matplotlib, scipy, yfinance

⚠ DISCLAIMER: This report is generated for educational and research purposes only. It does not constitute investment advice, solicitation, or an offer to buy or sell any securities. Past performance does not guarantee future results. All investments involve risk, including the possible loss of principal. Please consult a SEBI-registered investment advisor before making any investment decisions.

14. Public Glossary — Understanding the Metrics

Modern Portfolio Theory (MPT)

A mathematical framework for assembling a portfolio of assets such that the expected return is maximised for a given level of risk. It assumes that investors are risk-averse and that risk can be reduced through diversification.

Sharpe Ratio

The most widely used measure of risk-adjusted return. It tells you how much excess return you are receiving for the extra volatility you endure holding a riskier asset. A higher Sharpe ratio is better. (>1.0 is considered good, >2.0 is excellent).

Annualised Volatility

A statistical measure of the dispersion of returns, representing the 'risk' of the asset. It shows how much the asset's price fluctuates over a year. Lower volatility means smoother, more predictable returns.

Maximum Drawdown

The maximum observed loss from a peak to a trough of a portfolio, before a new peak is attained. It is a critical indicator of downside risk and helps investors understand the worst-case historical scenario.

Efficient Frontier

A graph representing a set of optimal portfolios that offer the highest expected return for a defined level of risk. Portfolios that lie below the efficient frontier are sub-optimal because they do not provide enough return for the level of risk.

Correlation Matrix

A table showing correlation coefficients between assets. A value of +1 implies they move perfectly together, 0 implies no relationship, and -1 implies they move perfectly in opposite directions. Combining assets with low or negative correlation reduces overall risk.

Monte Carlo Simulation

A computational technique that uses repeated random sampling to generate thousands of possible portfolio combinations. It helps map out the Efficient Frontier by testing a massive variety of weight allocations.

This glossary is provided to assist retail investors and the general public in interpreting institutional financial metrics.